

Figure 4. Distribution of high-confidence variants and ctDNA detectability before and after radiotherapy. (A) OncoPrint showing the distribution of high-confidence variants by patient and sampling timepoint. (B) Proportion of patients with detectable circulating tumour DNA (ctDNA) at baseline and post-radiotherapy. (C) Gene-level ctDNA detectability at each timepoint.

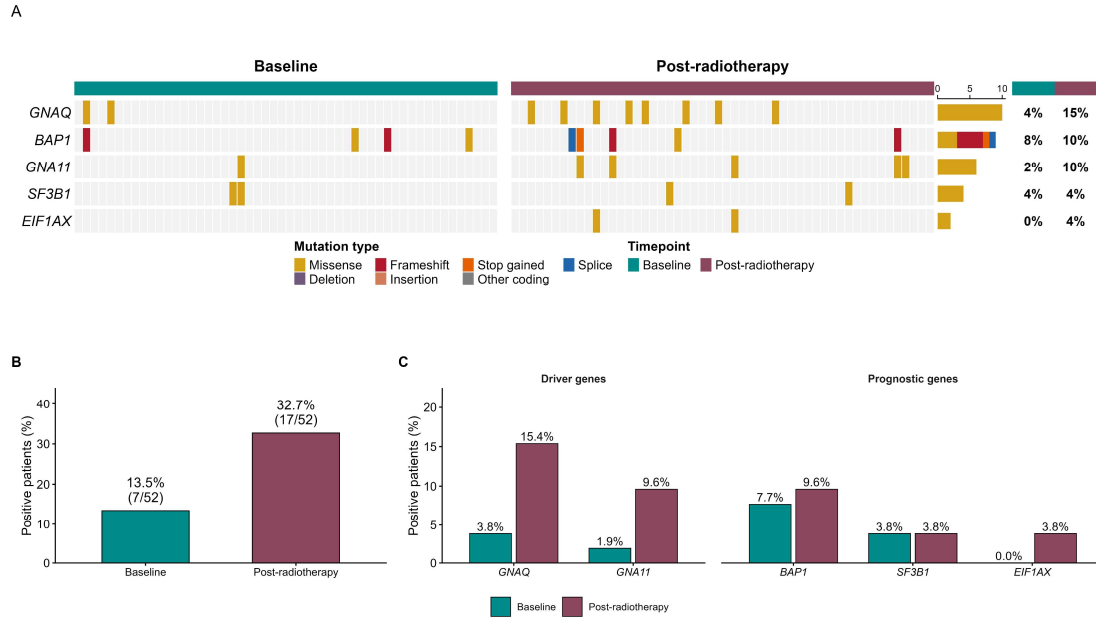


Table 1. Gene-level ctDNA detectability, variant distribution and maximum molecular VAF. Summary of circulating tumour DNA (ctDNA) detectability by gene, the distribution of high-confidence variants, and maximum molecular variant allele fraction (VAF) at baseline and post-radiotherapy.

Gene	Positive patients, n/N (%)			Variant distribution		Median maximum VAF (IQR), %	
	Baseline	Post-RT	Change (pp)	Variant calls BL / post	Unique variants BL / post	Baseline	Post-RT
Overall							
All genes	7/52 (13.5%)	17/52 (32.7%)	+19.2	9 / 22	8 / 13	0.077 (0.045-0.180)	0.084 (0.040-0.335)
Driver genes							
GNAQ	2/52 (3.8%)	8/52 (15.4%)	+11.5	2 / 8	1 / 2	0.047 (0.039-0.055)	0.062 (0.039-0.093)
GNA11	1/52 (1.9%)	5/52 (9.6%)	+7.7	1 / 5	1 / 2	0.056 (0.056-0.056)	0.099 (0.075-0.347)
Prognostic genes							
BAP1	4/52 (7.7%)	5/52 (9.6%)	+1.9	4 / 5	4 / 5	0.061 (0.045-0.096)	0.070 (0.052-0.335)
SF3B1	2/52 (3.8%)	2/52 (3.8%)	+0.0	2 / 2	2 / 2	0.445 (0.328-0.563)	0.393 (0.210-0.576)
EIF1AX	0/52 (0.0%)	2/52 (3.8%)	+3.8	0 / 2	0 / 2	Not detected	0.590 (0.354-0.826)

Detectability uses 52 paired patients at each endpoint. Maximum molecular VAF is summarized across positive patients only. Overall exact paired McNemar P=0.0309 (14 gained and 4 lost detection). No paired VAF-test was performed because only 3 patients were positive at both endpoints.